

Gamma detection for source localisation in targeted alpha therapy

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Targeted alpha therapies can deliver high localised dose, but treatment planning and delivery depend on whether the radionuclide source remains near the intended site after administration. In the AlphaGlue delivery concept [1], α -emitters are applied to a tumour region in a thin glue layer. Here we present a Geant4 simulation study showing that decay-chain gamma detection can distinguish retained, broadly diffused, and locally displaced source states in an AlphaGlue-like geometry. Using 50 independent repetitions, baseline-subtracted gamma-hit profiles reconstructed the broad diffusion fraction with $R^2 = 0.9987$ and about one percentage-point mean absolute error, while localised-source residuals identified displaced activity with mean position errors of 16–30 mm. These results support gamma detection as a candidate route for emitter-position monitoring, treatment planning, and delivery verification.

Introduction

In targeted alpha therapy, the delivered dose is strongly local. A treatment plan therefore depends not only on the initial activity distribution, but also on whether source material and progeny remain near the intended deposition site, diffuse into surrounding tissue, or are redistributed by biological transport [2, 3]. The AlphaGlue concept proposes suspending alpha emitters in a thin glue layer applied to tissue [1]. The concept has previously been evaluated with Geant4 and Geant4-DNA simulations for dose and DNA-damage modelling [4, 6]. This study investigates whether the accompanying gamma signal can be used as a theranostic observable, to monitor the emitter position, detect source movement, and provide useful information as a function of detection time.

Simulation model

Photon transport was simulated with the Geant4 toolkit [4]. The model places a 40 mm $\times$ 2 mm $\times$ 40 mm loaded glue layer on the $+y$ surface of a 20 cm $\times$ 20 cm $\times$ 150 cm water phantom. Six detector panels surround the phantom at 20 cm from the water surface. Each photon entry into a detector panel is counted as one hit.

Seven source configurations were studied to separate three clinically relevant possibilities: emitter retained in the loaded glue layer, emitter broadly mixed into the surrounding volume, and emitter transported to a localised displaced region. The retained-source case is used as the baseline. As discussed in [2, 3], some of the α -emitters in the decay chain can be taken up by the blood. [2, 3] report large variations in this uptake across different experiments, further emphasizing the need for local dose verification. This is taken into account in the

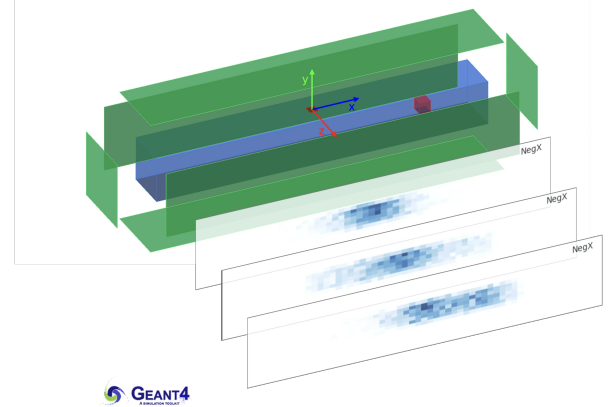


Figure 1: *Simulation geometry showing the water phantom (blue), surrounding detector panels (green), a displaced source (red), and example detector projections.*

simulation by studying cases where 10%, 50%, or 90% of the radioisotope is placed diffusely in the water volume, while the localised-displacement cases place 20%, 40%, or 60% of the source in a small displaced water region. These configurations are intended as sensitivity tests for source movement and source localisation.

Each configuration was simulated with 50 independent repetitions of 10,000 primary decays, giving 500,000 simulated decays per configuration and 3.5 million simulated decays across all source states. The repeated runs define the run-to-run uncertainty used in the residual and detectability analysis. Where activity-scaled observables are quoted, the simulated decays are rescaled to a DaRT-like reference activity of 3 μ Ci (111 kBq), matching the order of a single DaRT seed [5]. The simulated geometry is shown in Fig. 1.

Baseline subtraction

The total detector-entry rate changed only weakly between retained and redistributed source states. The total detector-entry rate changed by only about 3.8% between the retained baseline and the 90% broad-diffusion case. In contrast, baseline-subtracted profiles show a clear spatial signature. The retained source defines the reference profile $\bar{h}_p^B(z)$ for each detector panel p . Candidate source states are represented by residuals

$$\Delta h_p(z) = h_p(z) - \bar{h}_p^B(z),$$

with the repeated baseline simulations defining the bin-wise noise scale. Positive residuals mark detector regions where the candidate produces more gamma entries than the retained-source case; negative residuals mark depleted regions. Broad diffusion widens the residual pattern and changes the balance between the glue-facing and opposite detector panels, while localised displacement produces a shifted excess near the displaced source region. This indicates that source movement is better detected

from baseline-subtracted spatial hit patterns than from the total hit rate alone (Fig. 2).

Residual-based source metrics

The residual analysis was organised into three source-monitoring metrics. First, diffusion was inferred from profile-shape features and panel-balance changes. The $+y$ panel response decreases as activity moves away from the glue layer, while the profile width and the side-panel share increase. Second, activity-retention observables used the central $+y$ region of interest as an applicator-facing signal. This estimates activity still consistent with the retained-source geometry, but is interpreted only after a secondary-source test because compact displaced activity can also suppress the central signal (Fig. 3). Third, compact secondary sources were searched for using positive residuals in a displaced-region ROI and panel-asymmetry changes that are small for broad diffusion but large for off-applicator localisation.

For the present geometry, the displaced-source search ROI is centred on the positive- z region containing the simulated compact water source. The secondary-source score integrates the positive residual in this ROI and compares it with the retained-source baseline variation. If this test is positive, the location is estimated from the positive-residual centroid on the detector views that best project the source onto the longitudinal axis. This cascade is important: diffusion fraction is reported for broad redistribution candidates, while localised cases are first reported as compact secondary-source detections with an estimated position. The leakage comparison is shown in Fig. 4.

Signal inference

Using profile-shape features from repeated simulations, the broad diffusion fraction was recovered with $R^2 = 0.9987$ and a mean absolute error of about one percentage

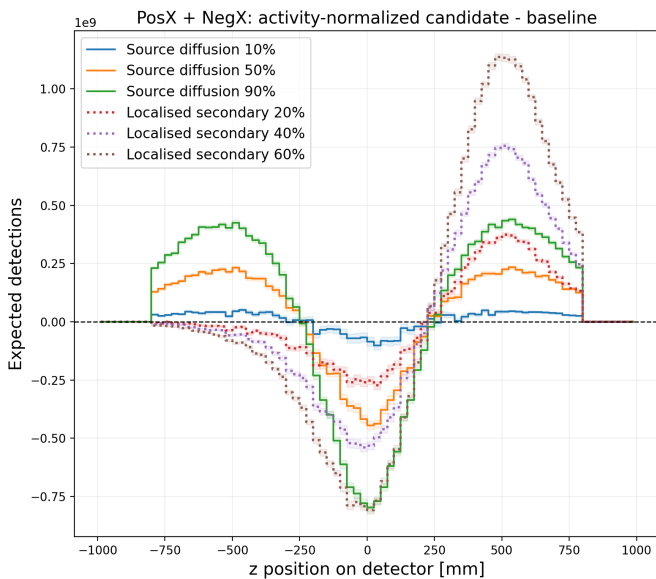


Figure 2: Activity-normalised, baseline-subtracted $+x$ and $-x$ detector profiles for broad diffusion and localised displacement. Shaded bands show variation across repetitions.

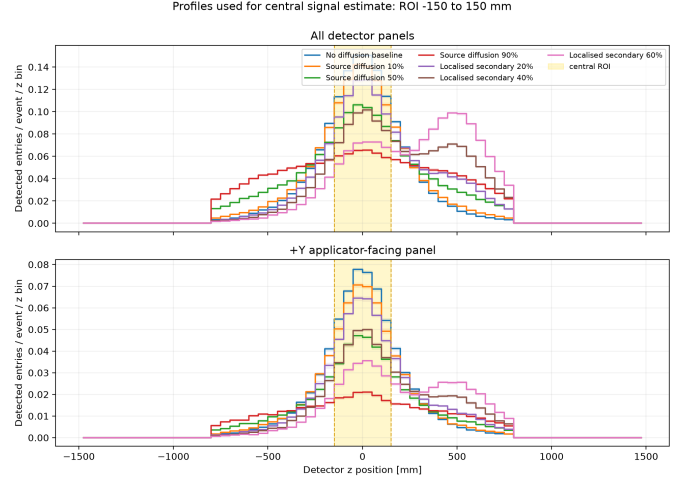


Figure 3: Central-ROI signal used to separate retained applicator response from activity redistributed outside the applicator region.

Table 1: Summary of first 3σ detection times from the time-window analysis of 50 independent repetitions.

Case group	First 3σ detection
10% broad diffusion	about 6 h
50–90% broad diffusion	about 1–2 h
20–60% localised displacement	about 1–2 h

point. A separate secondary-source metric identified the localised displaced source in all 20%, 40%, and 60% localised-displacement cases. The reconstructed displaced-source positions were approximately $z = 516$ mm, 528 mm, and 530 mm for 20%, 40%, and 60% localised activity, respectively, compared with the simulated $z = 500$ mm source region. Across these localised fractions, mean position errors were 16–30 mm (Figs. 5 and 6).

Time dependence

Cumulative time-window profiles show that the baseline-subtracted signal is already visible on hour-scale windows and becomes stronger over longer observation windows. Across the 50 repetitions, detectability required candidate-baseline separation greater than 3σ of the run-to-run baseline variation. The 50–90% broad-diffusion and 20–60% localised-displacement cases crossed threshold in about 1–2 hours, while the weaker 10% broad-diffusion case required about 6 hours. A representative time-window profile and a separate ROI significance trajectory are shown in Fig. 7. A $t < 10^6$ s analysis cut suppresses a late decay-chain tail while retaining the early signal relevant to the simulated source-localisation window.

Outlook

These results indicate that decay-chain gamma detection can provide source-location information that is not captured by total count rate alone. Further work will include detector response, backgrounds, and patient-scale geometries so that the activity-normalised profiles can be interpreted as experimentally measurable count rates.

References

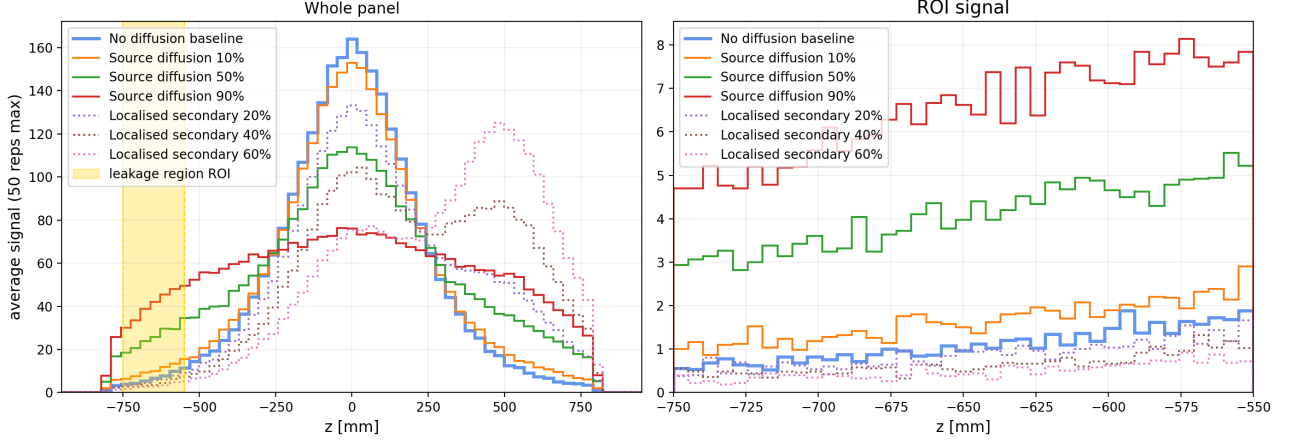


Figure 4: Whole x -panel profiles (left) and the displaced-region ROI signal (right). The ROI highlights the broad-diffusion excess outside the retained-source region.

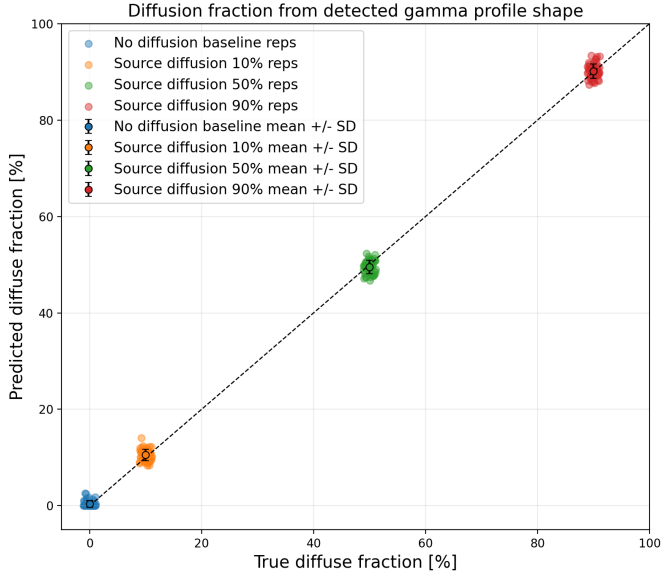


Figure 5: Diffusion fraction inferred from 50 independent repetitions; points show individual repetitions and large markers show the mean and standard deviation.

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Localised-source position residuals by source fraction

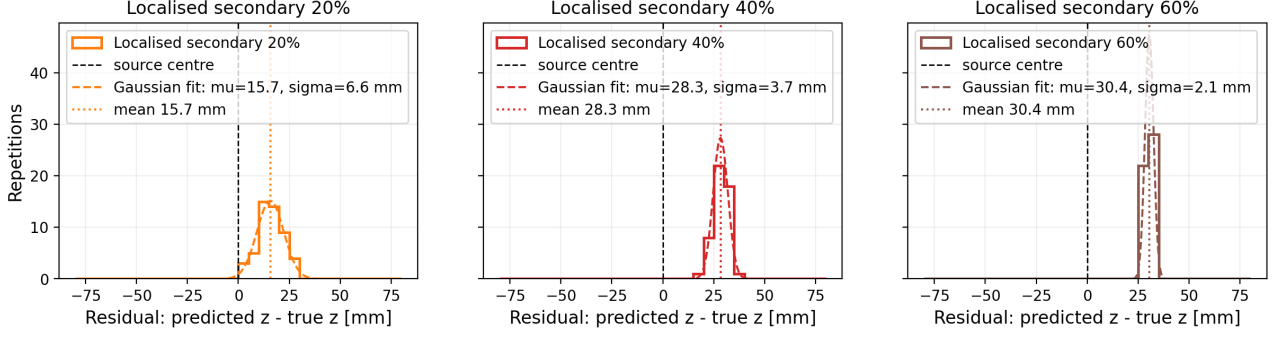


Figure 6: Localisation residuals across 50 repetitions for each displaced-source fraction. The dashed black line marks the true source centre; coloured lines indicate the mean and Gaussian fit.

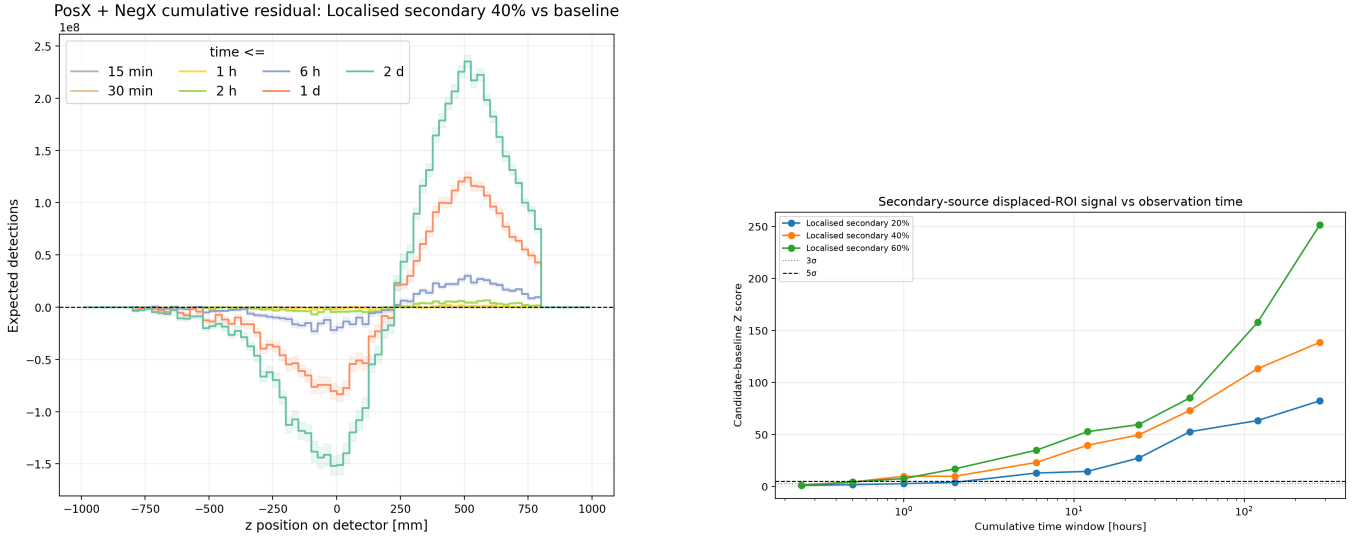


Figure 7: Time-dependent residual analysis. Left: the cumulative $+x$ and $-x$ profile for 40% localised displacement, showing a growing excess near $z = 500$ mm. Right: significance of the distinct displaced-source ROI signal with observation time.